THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS V — MATHEMATICS · MODEL PAPER 10 ANSWER KEY

Answers and marking points | Maximum Marks: 80

UNDERSTANDING SHAPE AND SPACE — Marks 21

Q.	Marks	Answer / marking points
1	1	Each square of the figure is of area 1 square cm. The area of the shaded figure is **(c) 19 sq cm**
2	6	**(i)** Name the type of each angle: (a) ∠K (b) ∠L (c) ∠M [3] (a) acute (b) obtuse (91° is a little more than a right angle) (c) reflex **(ii)** (a) Through how many degrees does the minute hand turn from … [3] (a) 9 × 30 = **270°** (b) **reflex** angle (more than 180°) (c) 180 ÷ 90 = **2**
3	6	**(i)** Name the flat shape of the face: (a) a face of a cube (b) th… [3] (a) square (b) circle (c) triangle **(ii)** Name the solid: (a) it has 2 flat circular faces and 1 curve… [3] (a) cylinder (b) cuboid (c) cone
4	8	The plan of a house in Bikaner is shown. The north direction is marked. **(a)** Find the area of the Hall. [2] 8 × 6 = **48 sq m** **(b)** Which room is to the south of the Study? What is its area? [2] The **Store**; area = 4 × 3 = **12 sq m** **(c)** Find the perimeter of the Study. [2] 2 × (4 + 6) = **20 m** **(d)** Find the area of the whole house. [2] Length 12 m, breadth 6 + 3 = 9 m; area = 12 × 9 = **108 sq m**

NUMBER SYSTEM — Marks 9

Q.	Marks	Answer / marking points
5	1	In 7,24,983, the digit in the lakhs place is **(b) 7**
6	1	3,48,562 rounded off to the nearest thousand is **(a) 3,49,000**
7	1	The number that comes just before 5,00,000 is **(c) 4,99,999**
8	6	**(i)** Rakesh lives 1 km from the bus stand. He walked 3/8 km and t… [3] (a) 3/8 + 1/4 = 3/8 + 2/8 = **5/8 km** (b) 1 − 5/8 = **3/8 km** (c) 3/8 of 1000 = **375 m** **(ii)** The line from 0.3 to 0.4 is divided into 10 equal parts. Wri… [3] A = 0.32 B = 0.35 C = 0.39

OPERATIONS — Marks 25

Q.	Marks	Answer / marking points
9	1	What should be added to 4,567 to get 10,000? **(b) 5,433**
10	3	A water tanker brings 3,600 litres of water to a village in Jaisalmer. Each of 150 familie… Water needed in one day = 150 × 12 = 1800 L Days = 3,600 ÷ 1800 = **2 days**
11	3	Divide 7,305 by 8. Find the quotient and the remainder. Verify the result. Quotient = **913**, remainder = **1**. Verification: 8 × 913 + 1 = 7304 + 1 = 7305. ✓
12	6	**(i)** Before Diwali, Mohan bought 12 packets of diyas at ₹ 35 per … [3] (a) diyas ₹ 420 + sweets ₹ 720 + crackers ₹ 365 = **₹ 1,505** (b) 2,000 − 1505 = **₹ 495** **(ii)** Find by a quick method: (a) 36 × 11 (b) 4 × 17 × 25 (c) 125 … [3] (a) 36 × 10 + 36 = 360 + 36 = **396** (b) (4 × 25) × 17 = 100 × 17 = **1,700** (c) 125 × 8 = 1000, so 125 × 16 = **2,000**

Q.	Marks	Answer / marking points
13	6	Bulbs numbered 1 to 40 are fixed in a row. Switch A lights every 8th bulb and Switch B lig… **(i)** (a) Write the numbers of the bulbs lit by Switch A. (b) Writ… [3] (a) **8, 16, 24, 32, 40** (b) **10, 20, 30, 40** (c) bulb **40**, the LCM of 8 and 10 **(ii)** (a) Find the LCM of 10 and 15. (b) Three bells ring every 10… [3] (a) **30** (b) LCM of 10, 15, 20 = 60 min, so at **7:00 pm** (c) **8**
14	6	**(i)** Look at the number grid. (a) Write the numbers that are mult… [3] (a) **9, 6, 3** (b) 4 + 6 + 8 = **18** (c) 9 − 1 = **8** **(ii)** 36 square tiles are arranged in a rectangle using all the ti… [3] Pairs: 1×36, 2×18, 3×12, 4×9, 6×6. (a) **5** (b) **6 × 6** (c) 2 × (9 + 4) = **26 m**

MEASUREMENT — Marks 17

Q.	Marks	Answer / marking points
15	3	Fill in the blanks: (a) 6 km 5 m = _____ m (b) 2 kg 500 g = _____ g (c) 5 hours 20 m… (a) 6005 (b) 2500 (c) 320
16	3	Write True or False: (a) 7:05 pm is written as 19:05 in the 24-hour clock. (b) ₹ 6.50 is t… (a) True (b) False (₹ 6.50 = 650 paise) (c) False (45°C is very hot)
17	3	A shopkeeper in Jaisalmer had 10 kg of dates. He sold 2 kg 750 g in the morning and 3 kg 4… Sold = 2750 g + 3480 g = 6230 g = 6 kg 230 g Left = 10,000 − 6230 = 3770 g = **3 kg 770 g**
18	8	The road map shows some cities of western Rajasthan with the distances in km. **(a)** Which is shorter for going from Bikaner to Jodhpur: the dire… [2] Through Nagaur: 110 + 135 = 245 km; direct = 250 km. Through **Nagaur** is shorter by **5 km**. **(b)** How many km more is it to go from Jaisalmer to Jodhpur throu… [2] 160 + 215 = 375 km; 375 − 285 = **90 km** **(c)** Which city is the nearest to Jodhpur on this map? Write the … [2] **Nagaur**, 135 km (Barmer is 215 km) **(d)** A bus goes from Bikaner to Jaisalmer and comes back by the s… [2] 2 × 330 = **660 km**

DATA HANDLING AND PATTERNS — Marks 8

Q.	Marks	Answer / marking points
19	1	Which of these numbers is both an even number and a square number? **(d) 36**
20	1	Squares are arranged in L-shapes as shown. How many squares are needed for Fig. 8? **(a) 15 — 1, 3, 5, 7, … (add 2 each time); Fig. 8 = 2 × 8 − 1**
21	3	The bar graph shows the number of books issued from a school library on 5 days. **(a)** On which day were the most books issued? [1] Thursday (30 books) **(b)** How many more books were issued on Tuesday than on Wednesday… [1] 24 − 12 = 12 **(c)** How many books were issued in the 5 days together? [1] 18 + 24 + 12 + 30 + 21 = 105
22	3	Look at the pattern: 1, 1, 2, 3, 5, 8, ___, ___. (a) What is the rule of the pattern? (b) … (a) each number is the sum of the two numbers before it (b) 5 + 8 = 13, 8 + 13 = 21 (c) 13 + 21 = 34, 21 + 34 = 55 → **55**

END OF ANSWER KEY — Total: 80 Marks